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A semi-Lagrangian semi-implicit immersed boundary method for atmospheric flow over complex terrain

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ABSTRACT

With the increasing resolution of numerical weather prediction models the slopes of orographic features such as hills and mountains is becoming much better resolved with the consequence that the gradients encountered are becoming ever steeper. The commonly used terrain following coordinate, which actually becomes singular at 90°, is becoming more problematic due to the resulting stiffness making the elliptic boundary value problem difficult to solve. Even when a solution is obtainable the feedback of the pressure correction onto the momentum will generally cause instabilities and/or unphysical solutions. This paper uses an alternative representation of the orography, namely the immersed boundary method, in a semi-implicit-semi-Lagrangian model of the inviscid compressible Euler equations. The results show that the method not only performs well but, due to the chosen method of implementation, that the computational overhead of the geometry can be decoupled from implicit treatment of the vertical gravity wave component.

1. Introduction

It is common in atmospheric models for the orography to be represented using a terrain following coordinate system such as that introduced in [1]. The primary advantage of this type of transformation is that the vertical coordinate remains aligned with the direction of gravity which reduces potential errors associated with this, rather large, force. This is especially important for large scale atmospheric motions where the atmosphere is in approximate hydrostatic balance. However with the increasing resolution of models, especially in the Limited-Area setting where only a small portion of the Earth is modelled, mountains and other steep orographic features are becoming better resolved and the deformation/non-orthogonality of the grid transformation makes the solution of the elliptic boundary problem for the pressure correction problematic due to large off-diagonal entries in the matrix. There has been some success at stabilizing this issue using a more complete matrix formulation with a suitable choice of preconditioner as shown in [2] (see also [3]) but the results of the simulations for slopes greater than around 60° show significant grid imprinting above the surface even though the model can run stably at even steeper slopes. Furthermore when stratification effects and compressibility are also present, as within the Met Offices ENDGame model [4], the ability to solve the nonlinear problem efficiently as well as the, potentially unrealistic, large amplitude (spurious) breaking gravity waves which can cause issues with the ellipticity of the Helmholtz problem and make the stable integration either problematic or impossible.

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Indeed it has been the authors experience that issues can arise at rather mild slopes, as low as 25° depending on the weather, which tends to cause unwanted failures in high resolution case/process studies such as flow over the Alps, where the terrain needs to be filtered sufficiently that the model can be run. This filtering will often reduce the orographic gradients by up to a factor of 2 with the result that the simulations produce significant errors in position and quantity of rain bands. There is thus a need to find an alternative method for representing flow over steep terrain for use in future models.

One possible remedy to this issue, which has the added advantage of retaining a simple grid structure that can accurately represent hydrostatic balance, is the Immersed Boundary Method (IBM) introduced by Peskin [5,6]. In this method the boundary is represented by point sources (discrete Dirac delta functions) whose value is to be determined by the requirement that the boundary conditions must hold, in a distributed sense, at the source points. In many ways this is not dissimilar to the method of Green's functions which the method reduces to for simple linear problems such as potential flow. A simple, informative, example of the basic ideas behind this method can be found in [7].

Since the original inception there have been many variations of the method proposed by [5] such as the ghost-cell method (see [8] for example) where rather than adding the sources explicitly to the continuous equations the discrete equations are manipulated near the boundary in order to satisfy the boundary conditions – the so-called direct method. An alternative to this is the indirect-direct method [9] which calculates a source by evaluating the equation at the boundary assuming the boundary condition holds. For a fuller review of these methods and further references see the review articles [10,6]. Although the family of direct methods, which are not dissimilar to the older cut-cell technique, appear to be better at representing the interface between the fluid and solid they suffer from the need to perform a lot of pre-processing and extraneous coding near the interface in order to apply the boundary conditions. Peskin's method [5,6] on the other hand is relatively straightforward when it comes to modifying the equations but suffers from the added complexity of the solution of the coupled system of equations for the pressure and sources. [3] applied a variant of the feedback control method of [11] for flow over urban type flows and compared the results to both wind tunnel data and a terrain following version of the model. In both these papers, although not a necessary requirement for the explicit form, the immersed boundary was essentially assumed to be on a grid point of the computational mesh [12]. Indeed [3] appears to force the velocity to zero within the immersed boundary. More recently [13] and [14] have implemented a version of the ghost-cell method into the Weather Research and Forecasting (WRF) model using a ghost-cell technique while [15] used the model of [16] to perform Large-eddy simulations over both an idealized and a real hill. The method of [9] has also been recently employed by [17] for high resolution simulations of turbulent flow over hills.

For the present problem it was decided to use Peskin's approach for simplicity although, in line with the direct method, the sources are not added directly to the full equations but to the semi-discrete equations after some manipulations have been performed. For isentropic flow this is actually still identical to Peskin's methodology but for stratified flow the sources are added after the rearrangement of the semi-implicit buoyancy terms which reduces the computational cost since the matrix representing the immersed sources becomes independent of time.

The present approach also differs in its application as well as the details of the numerics employed. Firstly, since the problem of interest is steep orography in an atmospheric model the equations are inviscid and compressible (see [18] and §2) and so the boundary conditions are simply a no-normal flow constraint rather than the viscous no-slip condition. Secondly since the concern is with simulations involving moderate to high Courant numbers (both flow and acoustic)[1] the model must be stable and relatively robust. For this reason, and the fact that the method must be applicable to the Met Office ENDGame model [4], a semi-implicit-semi-Lagrangian approach is employed with the boundary conditions enforced implicitly, to some convergence tolerance, on the nonlinear equations.

2. Model equations and discretisation

Let $(x_1, x_2, x_3) = (x, y, z)$ denote the usual Cartesian coordinates and $(u_1, u_2, u_3) = (u, v, w)$ the corresponding velocity components with respect to the usual Euclidean basis $(\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3) = (\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k})$. The compressible Euler equations are then

$$\frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} = -C_p \theta \nabla \Pi - \boldsymbol{g}, \tag{2.1}$$

$$\frac{\mathrm{D}\theta}{\mathrm{D}t} = 0, \tag{2.2}$$

$$\frac{\mathrm{D}\Pi}{\mathrm{D}t} = -\gamma^{-1} \Pi \nabla \cdot \boldsymbol{u}, \tag{2.3}$$

where, using the Einstein summation convention,

$$\frac{\mathrm{D}}{\mathrm{D}t} \equiv \frac{\partial}{\partial t} + u_k \frac{\partial}{\partial x_k} \tag{2.4}$$

[1] In a typical Limited area model used at the Met Office the vertical acoustic Courant number is $O(4000)$ while in the horizontal the flow Courant number can be > 2 and $O(10)$ for the sound waves.

is the usual material derivative following the fluid. Here θ is the potential temperature, Π is the Exner pressure and $\boldsymbol{g} = g\boldsymbol{k}$ with $g = 9.81$ [$\mathrm{ms^{-2}}$] is the acceleration due to gravity. Note that this formulation follows closely that of [18] and, in line with that paper, the equation for density has been eliminated via the equation of state for an ideal gas.

The Constants $\gamma = (1-\kappa)/\kappa$ and $\kappa = R/C_p$ are related to the specific gas constant $R = 287.05$ [$\mathrm{J\,kg^{-1}\,K^{-1}}$], and the specific heat capacity $C_p = 1005$ [$\mathrm{J\,kg^{-1}\,K^{-1}}$] while the Exner function is related to the pressure p, by the relation $\Pi = (p/p_0)^\kappa$ where $p_0 = 100000$ [Pa] is the mean surface pressure. Finally the temperature T is related to θ and Π by the simple relation $T = \Pi\theta$.

In the present work we are primarily interested in inviscid flows, hence the absence of viscous stresses in (2.1), and so we require the *no normal flow* condition

$$\boldsymbol{n} \cdot \boldsymbol{u} = 0, \tag{2.5}$$

on the immersed boundaries wherein $\boldsymbol{n}$ is the unit normal to said boundaries. This condition will be discussed in detail later when the immersed boundary sources are added to the equations.

2.1. Departure point algorithm

Since the underlying grid is Cartesian the characteristics of the operator (2.4) (i.e. the particle paths) satisfy the ordinary differential equations

$$\frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t} = \boldsymbol{u}. \tag{2.6}$$

This system of equations is solved using a second order Runge-Kutta scheme the first stage of which is simply the forward Euler step

$$\boldsymbol{x}|_d = \boldsymbol{x}|_a - \delta t \boldsymbol{u}^{n+1}|_a, \tag{2.7}$$

where δt is the length of the time step and the $|_d$ and $|_a$ are used to denote evaluation at the departure and arrival points respectively. The second order correction to this first order estimate is

$$\boldsymbol{x}|_d = \boldsymbol{x}|_a - \tfrac{1}{2}\delta t[\boldsymbol{u}^n|_d + \boldsymbol{u}^{n+1}|_a], \tag{2.8}$$

with the value of $\boldsymbol{u}^n|_d$ obtained by linearly interpolating the velocity vector to the previously predicted departure point. Note that $\boldsymbol{u}^{n+1}$ is not yet known and has to be found as part of an iterative process described later (but see [18] for more details).

2.2. Semi-discrete system

The equations (2.1), (2.2) and (2.3) are solved using a standard semi-implicit weighted-averaging approach which reduces to the classic Crank-Nicolson scheme when the off-centering parameter α is $1/2$ but typically a value slightly greater than this (0.51) will be used. In order to simplify the notation somewhat it is natural to introduce the time-level n sources which will need to be interpolated to the departure points:

$$\begin{aligned}
\boldsymbol{R}_u^n &= \boldsymbol{u}^n - (1-\alpha)\delta t[C_p\theta^n\nabla\Pi^n + \boldsymbol{g}], \\
R_\theta^n &= \theta^n, \\
R_\pi^n &= \Pi^n[1 - \gamma^{-1}(1-\alpha)\delta t\nabla\cdot\boldsymbol{u}^n].
\end{aligned} \tag{2.9}$$

The derivation up to this point has been semi-discrete in that no mention has been made of the discrete representation of the various spatial derivatives. The reason for this is that it complicates the notation and, to an extent, hides the simplicity of the mathematics which is independent of the choice of finite differences and variable staggering. The present model uses a simple Arakawa "C-grid" with the pressure (Exner) stored at cell centres and the normal components of the velocity on the cell boundaries. The thermodynamic variable θ is collocated with the vertical velocity, the so-called Charney-Phillips staggering, which is the same arrangement of variables employed by the current operational Met Office model ENDGame [4] (see also [18]). All spatial derivatives use the usual centered form with simple averaging employed for the evaluation of quantities at nearby grid points when required.

The nonlinear system of equations that needs to be solved can now be written as

$$\begin{aligned}
\boldsymbol{u}^{n+1} &= \boldsymbol{R}_u^n|_d + \boldsymbol{Q}_u^{n+1} = R_u^n|_d - \alpha\delta t[C_p\theta^{n+1}\nabla\Pi^{n+1} + \boldsymbol{g}], \\
\theta^{n+1} &= R_\theta^n|_d, \\
\Pi^{n+1} &= R_\pi^n|_d + Q_\pi^{n+1} = R_\pi^n|_d - \alpha\delta t\gamma^{-1}\Pi^{n+1}\nabla\cdot\boldsymbol{u}^{n+1},
\end{aligned} \tag{2.10}$$

where the subscripts a have been dropped on terms which are to be evaluated at the arrival point. Note that the above system of non-linear equations (2.10) will be solved using an approach closely related to a Newton-Raphson method with some approximations used to simplify the Jacobian matrix. The system of equations (2.10) are first recast into a defect correction form. This is achieved by making an estimate of a variable ψ^{n+1}, ψ^{n*} say, and forming an equation for the correction ψ' such that $\psi^{n+1} = \psi^{n*} + \psi'$ is a better approximation to the solution. The system (2.10) is now rewritten as, expanding the vectorial terms into components,

$$u' + \mathcal{H}_u^{n*}(\Pi') = \mathcal{R}_u = R_u^n|_d + Q_u^{n*} - u^{n*}, \tag{2.11}$$

$$v' + \mathcal{H}_v^{n*}(\Pi') = \mathcal{R}_v = R_v^n|_d + Q_v^{n*} - v^{n*}, \tag{2.12}$$

$$w' + \mathcal{H}_w^{n*}(\Pi') + H_b\theta' = \mathcal{R}_w = R_w^n|_d + Q_w^{n*} - w^{n*}, \tag{2.13}$$

$$\theta' + H_\theta^{n*} w' = \mathcal{R}_\theta = R_\theta^n|_d - \theta^{n*}, \tag{2.14}$$

$$D^{n*}\Pi' + H_\pi^{n*}\overline{w'} + \mathcal{H}_d^{n*}(\boldsymbol{u}') = \mathcal{R}_\pi = R_\pi^n|_d + Q_\pi^{n*} - \Pi^{n*}, \tag{2.15}$$

where the constants and operators are listed in Appendix A. This is essentially a simplified (Cartesian) form of the system of equations derived by the present authors in [18]. Note that the overbar in (2.15) denotes vertical averaging required by the grid staggering.

2.3. The unforced Helmholtz equations

In the following the superscripts $(n*)$ will be dropped from the operators/coefficients on the left-hand side of (2.11) to (2.15) in order to make the Helmholtz equation more readable. The first step is to eliminate θ' between (2.13) and (2.14) to obtain

$$w' + \hat{\mathcal{H}}_w(\Pi') = \hat{\mathcal{R}}_w = H_c(\mathcal{R}_w - H_b\mathcal{R}_\theta), \tag{2.16}$$

where $H_c = (1 - H_bH_\theta)^{-1}$ and $\hat{\mathcal{H}}_w = H_c\mathcal{H}_w$. Note that H_c must remain positive and so may need to be locally reset to prevent the loss of ellipticity of the Helmholtz problem when the term becomes too small or changes sign. Equations (2.11), (2.12) and (2.16) can be combined into the single vector equation

$$\boldsymbol{u}' + \boldsymbol{\mathcal{H}}_u(\Pi') = \boldsymbol{\mathcal{R}}_u. \tag{2.17}$$

This is the equation we modify to apply (2.5) on the immersed boundary. The reason we add the sources at this point, rather than say in (2.11), (2.12) and (2.13), is that it allows a decoupling of the H_c factor, and hence the current atmospheric thermodynamic state, from the immersed sources and thus renders the *immersed boundary condition matrix* (see below) independent of time. To conclude this section it is observed that (2.15) can now be written as:

$$\mathcal{L}(\Pi') \equiv D\Pi' - H_\pi\overline{\hat{\mathcal{H}}_w(\Pi')} - \mathcal{H}_d[\boldsymbol{\mathcal{H}}_u(\Pi')] = \mathcal{R}_\pi - \mathcal{H}_d(\boldsymbol{\mathcal{R}}_u) - H_\pi\overline{\hat{\mathcal{R}}_w}. \tag{2.18}$$

This is the Helmholtz equation which is to be solved in the absence of the immersed boundary sources which are the subject of the next section.

3. The immersed sources

Following the ideas of Peskin [6] we wish to modify equation (2.17) by adding a spatially varying force (integrated over the immersed boundary) which enforces the no-normal flow constraint (2.5) on the parametric surface $\boldsymbol{x} = \boldsymbol{X}(\sigma, \tau)$. Using distributions (2.5) can be replaced by

$$\int_{\mathcal{V}} \boldsymbol{u}\cdot\boldsymbol{n}\delta(\boldsymbol{x} - \boldsymbol{X}(\sigma,\tau))\mathrm{d}V = 0, \tag{3.1}$$

where $\delta(\cdot)$ is the Dirac delta function and $\mathcal{V}$ is the whole computational domain. Letting $\boldsymbol{X}_l$ and $\boldsymbol{n}_l$ $(l = 1, 2, \ldots, N)$ represent the discrete marker/sampling points of the surface and normal allows this boundary condition to be written, for each l, as

$$\sum_I \boldsymbol{u}_I\cdot\boldsymbol{n}_l\delta_h(\boldsymbol{x}_I - \boldsymbol{X}_l) \equiv \sum_I \boldsymbol{u}_I\cdot\boldsymbol{d}_{Il} = 0, \tag{3.2}$$

where I represents the set of all grid-points in the domain, $\delta_h(\cdot)$ is a discrete representation of the Dirac delta function (see [7] and Appendix C) and, trivially, $\boldsymbol{d}_{Il} = \boldsymbol{n}_l\delta_h(\boldsymbol{x}_I - \boldsymbol{X}_l)$. Note that, since $\delta_h(\boldsymbol{x}_I - \boldsymbol{X}_l)$ will be non-zero only on a small subset, I_l say, of the points of I means that the sum over I in (3.2) and subsequently can be replaced by one over I_l. Technically the sum in (3.2) should also be split since, due to the C-grid staggering, the different terms are to be evaluated at different locations on the grid. This slight abuse of notation however makes the formalism slightly more transparent than the more formerly correct one.

Since the velocity normal to the orography needs to be reduced to zero it is natural to add a (surface integrated) force to (2.17) which is normal to the marker. In continuous form this force would take the form of the surface integral

$$\boldsymbol{F}(\boldsymbol{x}) = -\int_{\mathcal{S}} F(\sigma,\tau)\boldsymbol{n}\delta(\boldsymbol{x}-\boldsymbol{X})\mathrm{d}\tau\mathrm{d}\sigma. \tag{3.3}$$

At this point it is worth noting, as mentioned in the introduction, that there is a strong connection with this methodology and the theory of Green's functions. In the 2D incompressible irrotational case or, via a Prandtl-Glauert transformation [19], in the subsonic problem the flow could be solved using contour integration in the complex plane for which the complex velocity potential will exhibit a jump of the residue of the circulation (via The Plemelj formula [20]) as a point crosses the contour (vortex sheet) of integration. This jump when differentiated will take the form of a Dirac delta function and is in line which the discussion of this section. In this idealized steady irrotational case the problem can actually be reduced to solving an integro-differential equation (Prandtl's equation [19,20]) for the circulation only which is similar to finding a single equation for the F in (3.3).

Using the discrete version of (3.3) equation (2.17) is thus replaced by

$$\boldsymbol{u}' + \boldsymbol{\mathcal{H}}_u(\Pi') = \boldsymbol{\mathcal{R}}_u - \sum_{l=1}^{N} F_l^{n+1}\delta_h(\boldsymbol{x}-\boldsymbol{X}_l)\boldsymbol{n}_l, \tag{3.4}$$

where the subscripts I have been dropped for convenience. Note also that the metric factors involving areas and volumes have been absorbed/canceled in this formulation. Recalling that $\boldsymbol{u}^{n+1} = \boldsymbol{u}^{n*} + \boldsymbol{u}'$ gives, in (3.2), for $l = 1, 2, \ldots, N$

$$\sum_I \boldsymbol{u}_I' \cdot \boldsymbol{d}_{Il} = -\sum_I \boldsymbol{u}_I^{n*} \cdot \boldsymbol{d}_{Il}, \tag{3.5}$$

which is the required boundary condition on the velocity correction.

It is natural, as with the other variables, to split the sources as $F_l^{n+1} = F_l^{n*} + F_l'$ with F_l^{n*} chosen, for each $l = 1, 2, \ldots, N$, such that

$$\sum_{m=1}^{N} B_{lm}F_m^{n*} \equiv \sum_{m=1}^{N}\left(\sum_I \boldsymbol{d}_{Im}\cdot\boldsymbol{d}_{Il}\right)F_k^{n*} = \sum_I (\boldsymbol{u}^{n*} + \boldsymbol{\mathcal{R}}_u)\cdot\boldsymbol{d}_{Il}. \tag{3.6}$$

From the above it should be evident that B_{lk} ($\boldsymbol{B}$) are the components of a sparse, symmetric positive definite matrix. Assuming a solution to this equation has been obtained then the sources $\boldsymbol{\mathcal{R}}_u$ can be updated and used in the Helmholtz equation (2.18). For later use this equation is written in the more compact form

$$\boldsymbol{B}\boldsymbol{f}^{n*} = \boldsymbol{b}. \tag{3.7}$$

The equation coupling the point sources F_l' and pressure correction Π' ((3.4), (3.5) and (3.6)) is now simply, for $l = 1, 2, \ldots, N$,

$$\sum_{m=1}^{N} B_{lm}F_m' + \sum_I \boldsymbol{\mathcal{H}}_u(\Pi')\cdot\boldsymbol{d}_{Im} = 0. \tag{3.8}$$

As already mentioned the present formulation has resulted in a matrix $\boldsymbol{B}$ which is independent of time and, more importantly, the atmospheric state. This not only means that $\boldsymbol{B}$ is computed once at model start-up but also approximate preconditioners/solvers (see below) can be built once and do not need to be recomputed. This essentially means that the rather costly construction of $\boldsymbol{B}$ and its approximation is not required during the time stepping of the model. In contrast, if the sources had been added to the equations before the manipulations resulting in (2.16), then $\boldsymbol{B}$ would need to be recalculated, possibly multiple times (H_c varies within the model iteration), per time step.

If the vector of unknown Exner values is denoted by $\boldsymbol{x}$ and the discretised Helmholtz operator $\mathcal{L}$ by the matrix $\boldsymbol{A}$ then the unforced problem (2.18) can be written as

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{r}, \tag{3.9}$$

where $\boldsymbol{r}$ is the discrete form of the right-hand-side of equation (2.18). Similarly if the vector of sources F_k' is denoted by $\boldsymbol{f}$ then (3.8) is of the form

$$\boldsymbol{B}\boldsymbol{f} + \boldsymbol{C}\boldsymbol{x} = \boldsymbol{0}, \tag{3.10}$$

where $\boldsymbol{C}$ is the non-square matrix representing the coupling terms $\mathcal{H}_u(\Pi') \cdot \boldsymbol{d}_{II}$; i.e. the contraction of the pressure gradient onto the immersed boundary. The final term to be discussed is the *spreading* operation $\boldsymbol{S}$ which takes the sources and distributes them to the velocity points and then contracts, through the divergence operator (see Appendix B for a more detailed description), onto the pressure grid. Combining this term with (3.9) and (3.10) yields the augmented problem

$$\begin{pmatrix} \boldsymbol{A} & \boldsymbol{S} \\ \boldsymbol{C} & \boldsymbol{B} \end{pmatrix} \begin{pmatrix} \boldsymbol{x} \\ \boldsymbol{f} \end{pmatrix} = \begin{pmatrix} \boldsymbol{r} \\ \boldsymbol{0} \end{pmatrix}. \tag{3.11}$$

This system of equations can be solved iteratively as follows, assume there are approximate factorizations

$$\boldsymbol{A} \approx \boldsymbol{LU}, \quad \text{and} \quad \boldsymbol{B} \approx \boldsymbol{GG}^T, \tag{3.12}$$

where $\boldsymbol{U}$ is an upper triangular matrix and $\boldsymbol{L}$, $\boldsymbol{G}$ are lower triangular matrices. In the present work the factors $\boldsymbol{L}$ and $\boldsymbol{U}$ are obtained via Stone's Strongly Implicit procedure (SIP(α), see [21]) with the parameter $\alpha = 0.9$ while the matrix $\boldsymbol{G}$ is derived from an incomplete block Cholesky factorization of $\boldsymbol{B}$ (see below for a further discussion). Using these approximations the fixed point iteration method

$$\begin{aligned} \boldsymbol{x}^k &= \boldsymbol{x}^{k-1} + (\boldsymbol{LU})^{-1}[\boldsymbol{r} - \boldsymbol{Sf}^{k-1} - \boldsymbol{Ax}^{k-1}] \\ \boldsymbol{f}^k &= \boldsymbol{f}^{k-1} - (\boldsymbol{GG}^T)^{-1}[\boldsymbol{Bf}^{k-1} + \boldsymbol{Cx}^k], \end{aligned} \tag{3.13}$$

for $k = 1, 2, \ldots$ can be obtained. This system is iterated until the normalized residual (L^2)

$$\frac{||\boldsymbol{r} - \boldsymbol{Sf}^{k-1} - \boldsymbol{Ax}^{k-1}||}{||\boldsymbol{r}||} \tag{3.14}$$

is less than some prescribed tolerance ϵ. In some cases, where the stiffness of the system (see next section) is slowing down the convergence, there is the option to replace (3.13) with

$$\boldsymbol{Ax}^k = \boldsymbol{r} - \boldsymbol{Sf}^{k-1}, \qquad \boldsymbol{Bf}^k = -\boldsymbol{Cx}^k, \tag{3.15}$$

for $k = 1, 2, \ldots$. The matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ are then approximately inverted using a restarted GCR(k) algorithm for $\boldsymbol{A}$ [22] and a preconditioned conjugate gradient method for $\boldsymbol{B}$ [23]. In the present implementation, when the Helmholtz matrix is solved using the Krylov methods, the incomplete factorizations described previously are used as preconditioners. These matrix inversions are solved to some prescribed (relative) tolerance of 10^{-4} although the block Cholesky factorization can be machine precision for sufficiently small problems since a full Cholesky factorization is then an option. It is important to note that the matrix $\boldsymbol{B}$ and its preconditioner can be pre-computed from the geometry of the grid and the immersed boundary locations at the beginning of the simulation and never need to be recalculated even though the Helmholtz matrix $\boldsymbol{A}$ and its preconditioner need to be due to the time varying nature of the coefficients in (2.18). An outline of the algorithm for performing a single time step using (3.13) is giving in Algorithm 1 which is a modification of that used in [18].

3.1. The immersed boundary grid, block factorization and stiffness

Currently the immersed boundary objects are created by joining together simpler geometric objects as illustrated in Fig. 1. For example the cube (and hemisphere) are generated from five 2D Cartesian grids by specifying the vertices, and unit normals, to each of the faces. The immersed boundary point is then placed at the centre of this quadrilateral with the normal vector calculated from the average of the vertices. Each of these blocks is treated as a separate entity when constructing the preconditioner so that the Cholesky factorization only needs to be performed on a relatively small sub-matrix. For example in Fig. 1 there are 23 pieces/blocks making up the 5 immersed objects. The full matrix $\boldsymbol{B}$, due to the unknown nature of the band structure, is stored in a *symmetric row compressed* format to reduce both the memory storage and number of floating point operations when performing matrix vector multiplications – in the example figure this matrix would be 5750×5750 with the vast majority of entries being zero and approximately half the non-zero entries stored twice due to symmetry.

As mentioned in the introduction there is a consensus in the literature that the application of the immersed boundary method to problems involving rigid boundaries, as is of interest here, that the resulting problem tends to be stiff which results in relatively small time steps in viscous flows. Although the current interest here is with essentially inviscid problems there also appears to be issues with the stiffness in the semi-implicit solver and so the origin of these issues is worth a short discussion. The first thing to note is that, when the immersed grid is higher resolution than the grid upon which the simulation is to be performed then the number of nonzero entries in each row increases due to the number of interactions between the immersed sources increasing (the overlapping of discrete delta functions). When this happens the matrix appears to become highly non-diagonally dominant which makes the matrix $\boldsymbol{B}$ very difficult to invert numerically in 3D (in 2D these issues are much milder) and, as is common in incomplete factorizations, the block Cholesky can fail. This numerical stiffness reduces drastically, and diagonal dominance is regained, when some care is taking to set the spacing between the immersed mesh points compared to the simulations grid. Specifically when the source spacing is comparable

Algorithm 1: Outline of the solution procedure for a single time step.

Input: Model state $(u^n, v^n, w^n, \theta^n, \Pi^n)$.
Data: Number of iterations per time step (*NumIts*; default is 2) of nonlinear system and solver tolerance (ϵ).
Set $(u^{n+1}, v^{n+1}, w^{n+1}, \Pi^{n+1}, \theta^{n+1}) = (u^n, v^n, w^n, \Pi^n, \theta^n)$;
Initialize fields (2.9) ready for interpolation to Departure points;
for *iter* = *1, ..., NumIts* **do**
  Estimate of departure points using (2.7) and (2.8);
  Interpolate sources (2.9) to departure points for (2.10);
  Update $\theta^{n+1} = R^n_\theta|_d$;
  Semi-implicit solver:
  begin
    Calculate the residual error terms in (2.11)-(2.15);
    Solve (3.6) using a preconditioner conjugate gradient method;
    Update source terms and build $\boldsymbol{r}$ for equation (3.9);
    Set initial guesses $\boldsymbol{x}^0$ and $\boldsymbol{f}^0$ to zero;
    $k = 0$;
    while *error* (3.14) $> \epsilon$ **do**
      Calculate $\boldsymbol{r}^k = \boldsymbol{r} - \boldsymbol{S}\boldsymbol{f}^k - \boldsymbol{A}\boldsymbol{x}^k$;
      Update guess $(3.13)_1$: $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + (\boldsymbol{LU})^{-1}\boldsymbol{r}^k$;
      Calculate $\boldsymbol{s}^k = \boldsymbol{B}\boldsymbol{f}^k + \boldsymbol{C}\boldsymbol{x}^k$;
      Update guess $(3.13)_2$: $\boldsymbol{f}^{k+1} = \boldsymbol{f}^k - (\boldsymbol{G}\boldsymbol{G}^T)^{-1}\boldsymbol{s}^k$;
      $k = k + 1$;
    end
  Using Π', update u', v' and w' using (2.11), (2.12) and (2.16);
  Calculate θ' using (2.14);
  Update all $(n+1)$ time level variables: $u^{n+1} \to u^{n+1} + u'$, etc.;
end

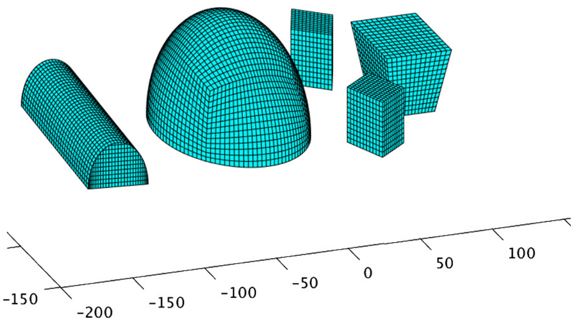


Fig. 1. Example of the immersed boundary grid.

to the grid then number of non-zero rows approaches 8 which is also almost the ideal spacing for linearly interpolating the C-grid velocities to a pressure point, which would be 6.

In a future development of this model we will likely create the grid at much higher resolution and then systematically discard problematic points and thus loose the regularity of the immersed grid. This is a viable option since the structured grid is only used for the generation of the immersed points and plotting and is not required (or currently used) by the main code since the immersed grid is stored in an indirectly addressed list. This means that some off-line grid and boundary matrix $\boldsymbol{B}$ optimization should be possible.

4. Results

As a first example of the method described we use a standard (2D) test case suggested in [24] and used in numerous papers such as [25] (and references therein) as a validation of numerical methods using boundary fitted coordinates. This test considers the flow over a 10% high bump which is a segment of a circle whose chord (distance between the points where it intersects the wall) is 100 m. The lid is then at a height of 100 m while the length of the domain is 300 m. Nondimensionally this corresponds to the domain described in [25]. Since the model currently employs periodic boundary conditions a *Rayleigh damping* region was inserted near the left side of the domain to relax the model state back to the balanced initial condition. This damping acts to return the outflow back to the inflow state smoothly over 25 grid points. The initial state is of an isentropic atmosphere with $\theta = T_0 = 288$ K and the surface value of Exner pressure $\Pi = 1$ (i.e. 10^5 Pa). For the free-stream velocity $w = 0$ and $u = U_\infty$ is constant and gravity is set to zero. The grid was set to be uniform with a spacing of 1 m in both directions which results in 300×100 points and the cross-stream derivatives are set to zero by ensuring the y halos (periodic wrap points) are equal to the centre line values. The distance between the immersed

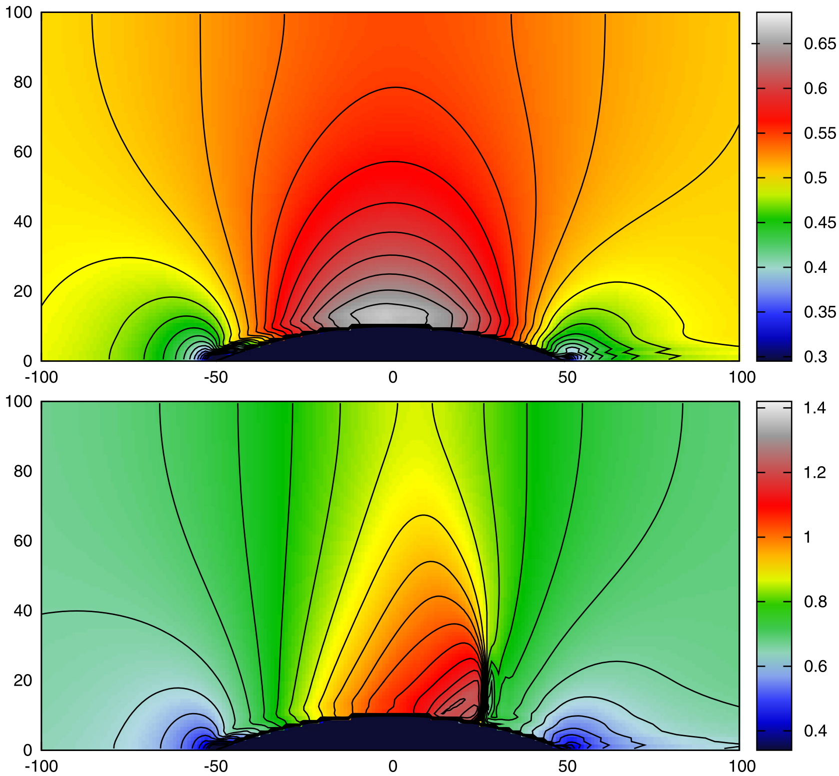


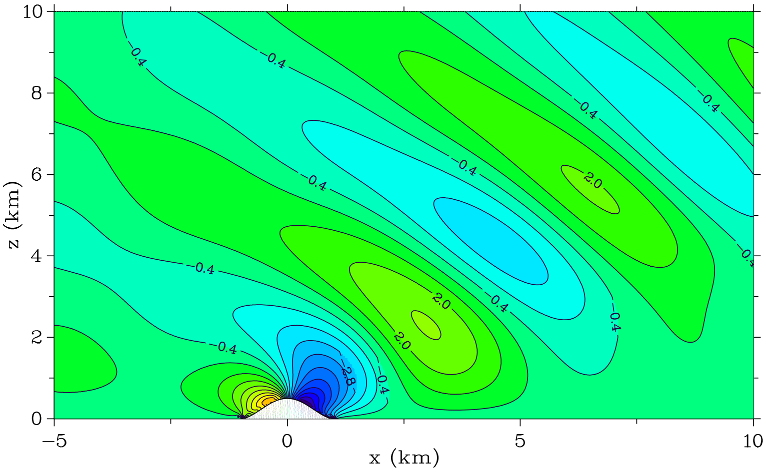
Fig. 2. Mach number contours for flow over a bump with mean Mach numbers of 0.5 (top) and 0.675 (bottom). (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

boundary points is constant and approximately 0.8 m. Fig. 2 shows the results from two different values of U_∞ namely $U_\infty = 0.5c$ (subsonic) and $U_\infty = 0.675c$ (transonic) cases. Here $c = \sqrt{C_p T_0/\gamma}$ is the speed of sound. Both simulations were run out until a steady state was reached with a *flow* Courant number of around 0.5 ($\delta t = 0.0025$ s and 0.002 s respectively). Although the wind speeds in this test are rather excessive for the types of flow of interest in meteorology it is encouraging that the present model reproduces this simple test so well – the contour levels chosen for Fig. 2 are exactly those used in [25]. There is a certain amount of asymmetry in these results as one may expect since the advection terms are all up-winded which introduces a certain amount of dissipation within the numerics which, like viscosity in a real fluid, acts to further slow the flow down on the leeward side of the bump. This is compounded by the fact that on the windward side transport is *onto* the boundary while on the leeward side it is away from the surface. Furthermore, since the method employed here is one of a diffuse boundary method rather than a so-called sharp interface method the effect of the points within the surface are further spread out due to the staggering of the grid; i.e. in order to plot the Mach number contours both velocity components need to be averaged to a common pressure point. It should also be pointed out that, since the surface is being modelled by a immovable vortex sheet, it is expected that although the normal velocity component will be continuous the tangential one should exhibit a jump discontinuity. This, as already noted, is in line with the 2D subsonic theory where the complex velocity potential will exhibit a jump of the residue of the circulation (via The Plemelj formula [20]) as a point crosses the contour of integration. This jump when differentiated will take the form of a Dirac delta function as discussed in §3; i.e. equation (3.3).

A more meteorologically interesting test is of non-hydrostatic nonlinear mountain waves. Here the atmosphere is assumed to be isothermal with a temperature of $T = 288$ K. The hill shape is described by the polynomial

$$f(x) = \begin{cases} h[1-(x/a)^2]^2, & |x| \le a, \\ 0, & |x| > a \end{cases} \tag{4.1}$$

where the hill height $h = 500$ m and half-width $a = 1000$ m. The flow is assumed to be at a Froude number of 1.5 which requires a free-stream velocity of 13.67 ms^{-1}. This is a particular difficult problem for the IBM method since although the hill is relatively small the scale of the downstream lee wave pattern is relatively large. In order to reduce reflections from the boundaries the lid was placed at a height of 25 km and the width of the domain was 30 km with 500 points in the vertical and 600 in the horizontal. Fig. 3 shows the vertical velocity component after 1500 timesteps ($\delta t = 1$ s). Also shown in Fig. 3 is the equivalent simulation using a terrain following model. As can be seen, even though the flow is still developing, the results are in rather good agreement.



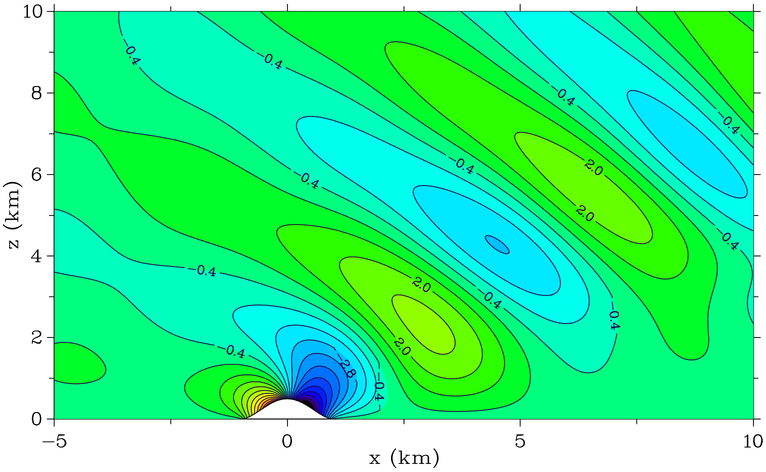


Fig. 3. Vertical velocity w for stably stratified flow over an idealized mountain. Top shows the results from the IBM model while bottom is the same simulation using a terrain following model.

As a better test of the application of the boundary conditions, as well as steep orography, is a modified version of the gravity current problem [26]. Here the background state is again isentropic ($\theta = 300$ K) with the atmosphere initially hydrostatically balanced and at rest. A temperature anomaly

$$\Delta T = \begin{cases} -15[1+\cos(\pi r)]/2, & R \le 1, \\ 0, & R > 1, \end{cases} \tag{4.2}$$

where ($r > 0$)

$$r^2 = \left(\frac{x - x_0}{a}\right)^2 + \left(\frac{z - z_0}{b}\right)^2, \tag{4.3}$$

is added to θ with $x_0 = 0$ and $z_0 = 3000$ m defining the centre of the perturbation and $a = 4000$ m and $b = 2000$ m are the horizontal and vertical lengths of the bubble. The domain for this problem is 25.6 km (length) by 6.4 km (depth) with a uniform grid spacing of 50 m and a time step of 0.5 s. The problem is modified by inserting, directly below the temperature anomaly, a semi-circular hill of radius of 1 km. Contours of the potential temperature at various times is shown in Fig. 4 as the bubble falls and encounters the immersed boundary. These results show how the density current is moved around the obstacle with very little penetration of the temperature into the obstacle. This is an interesting test since not only is θ an advected quantity which should remain constant on a particle path but it is also dynamically active since it is generating the fluid motion. This, along with the previous lee wave example, also demonstrates the effectiveness of the formulation in §3 in decoupling the (unsteady) buoyancy terms from the immersed boundary conditions since (1) the not insignificant cost of rebuilding the immersed boundary matrix at every iteration (H_c in (2.16) changes at every update by the semi-Lagrangian advection) has been avoided but also (2) the nonlinear iteration is converging to the no-normal flow boundary condition (in a distributed sense) as required. Although, in this example, the tracer does not appear to be penetrating the boundary

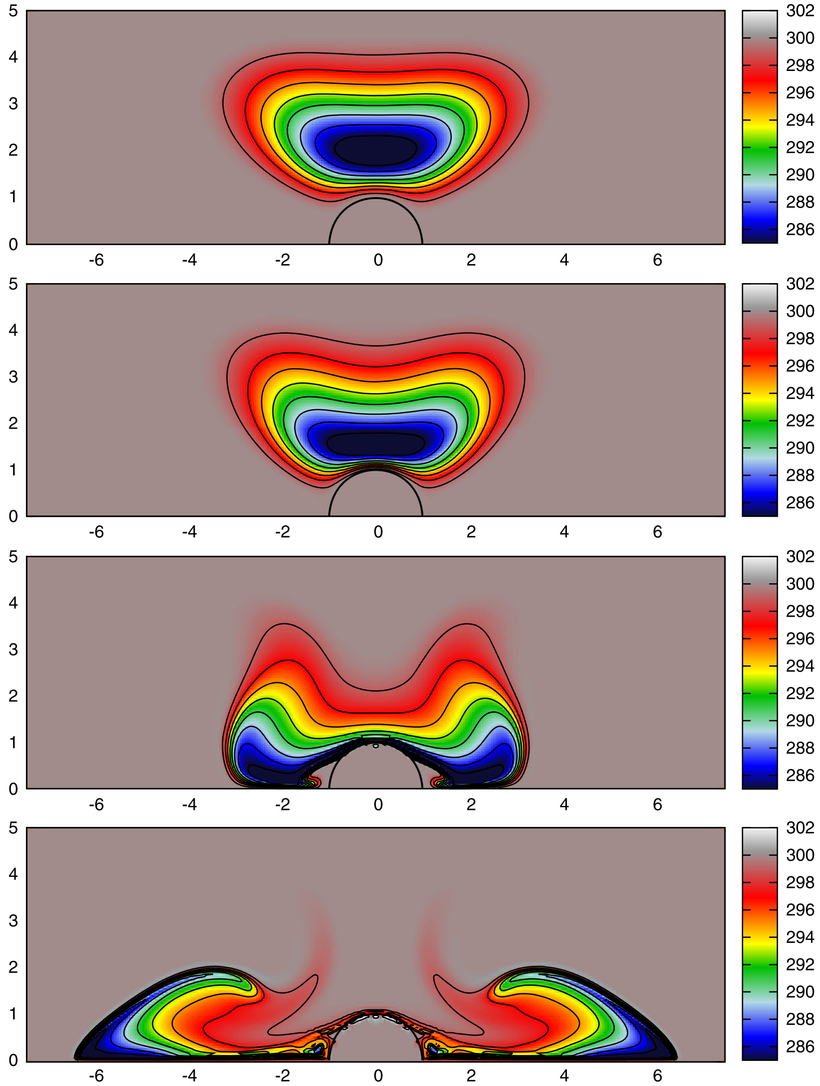


Fig. 4. Straka gravity current test over a semi-circular cylinder at different times. Top to bottom $t = 120$ s, $t = 150$ s, $t = 240$ s and $t = 360$ s. Note that the axes are in km.

it should be noted that even if there was θ perturbations within the obstacle it is not clear that this could be interpreted as *leakage* simply because of the nature of semi-Lagrangian advection. For example if at some grid cell inside the body the departure point lies sufficiently close to the boundary then the currently employed cubic Lagrange polynomial (a 4×4 stencil in 2D) is likely going to pick up values from outside the obstacle. If there is any loss or gain in the mass for the distinct regions then it is mainly due this non-conservative property of the interpolating semi-Lagrangian schemes (but see [27]). If conservation were required one could easily incorporate the volume-fraction semi-Lagrangian approach described in [28] to conserve the appropriate mass within each region.

As a final test of the model we consider a more elaborate (similar to an urban flow) problem based upon the grid shown in Fig. 1. Here the base state is isentropic with $\theta = 288$ K and the background wind is $u = 10$ ms^{-1} and $v = w = 0$. The domain is $200 \times 120 \times 60$ points in the streamwise, cross-stream and vertical directions respectively with a uniform grid spacing of 2.5 m in all directions. The time step is 0.1 s which results in a free-stream (flow) Courant number of 0.4. Near the inlet a damping layer is used to relax the flow back to the base state as was used in the 2D example of flow over a bump. To help visualize the flow a passive tracer is introduced, stored on pressure levels, which is initially zero. To simulate the behaviour of an inlet condition, as with the other variables, the tracer is relaxed to a prescribed profile within the damping zone. The specific profile chosen for the tracer is simply a linear function of height so that, for an undisturbed flow, the isosurfaces tend to planes parallel to the surface $z = 0$.

Fig. 5 shows the tracer after 25 seconds of simulation at which point the tracer has reached the top of the hemisphere. As can be seen the flow is correctly moving around/over the obstacle with very little penetration. Fig. 6 shows the tracer at a later time, $t = 80$ s, where it has had time to pass through the whole computational domain and, specifically, encountered

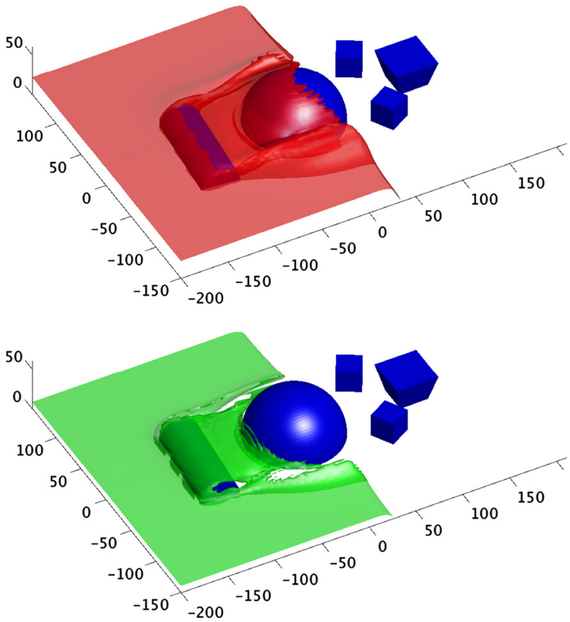


Fig. 5. 3d flow over complex geometry after 25 seconds. Top shows the isosurface for the tracer value of 0.6, while the bottom is for the isosurface value of 0.8.

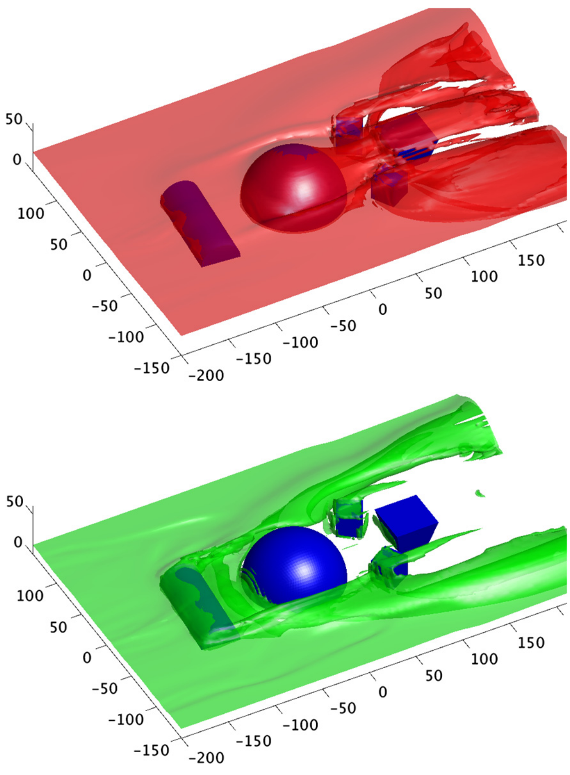


Fig. 6. Same as Fig. 5 but at a later time (80 s) after the tracer has had time to flow through the whole domain.

all of the immersed objects. Both figures show that, and as expected, the tracer moves around the objects and through the various gaps as dictated by the flow, without spuriously penetrating the surface of the obstacles, despite there being no physical solid boundaries present.

5. Conclusions

In this paper an initial investigation of the potential use of the immersed boundary method for high resolution regional (limited area) models has been performed using a semi-implicit semi-Lagrangian compressible flow model. The results show that the method is quite viable and robust despite complicated boundaries with more than 90° slopes. The additional computational cost is also not prohibitively expensive as shown by the fact that complex 3D test was performed on a single core of a desktop computer. Furthermore the flexibility and simplicity of the method make it feasible to combine it with a standard terrain-following approach, where the terrain following approach treats the smoother (non-steep) parts of the surface while the immersed boundary method takes care of the smaller fraction of the orography which is steeper. From an implementation point of view, not withstanding changes to the Helmholtz problem where most of the model changes occur, the present approach has the benefit of only requiring a list of data points and their normals. This makes it relatively straightforward to input the data into an existing model such ENDGame [4]. In contrast a ghost-cell method or variant would require substantial preprocessing in order to calculate the intersections of the body with the grid.

Appendix A. Helmholtz operators and coefficients

The Differential operators used in the Helmholtz equation (2.11)-(2.15) are:

$$[\mathcal{H}_u^{n*}(\Pi'), \mathcal{H}_v^{n*}(\Pi'), \mathcal{H}_w^{n*}(\Pi')] = \alpha\delta t C_p \theta^{n*}\left[\frac{\partial \Pi'}{\partial x}, \frac{\partial \Pi'}{\partial y}, \frac{\partial \Pi'}{\partial z}\right] \tag{A.1}$$

$$\mathcal{H}_d^{n*}(\boldsymbol{u}') = \frac{\alpha\delta t \Pi^{n*}}{\gamma}\left[\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} + \frac{\partial w'}{\partial z}\right]. \tag{A.2}$$

The remaining factors in (2.18) are:

$$H_b^{n*} = \alpha\delta t C_p \frac{\partial \Pi^{n*}}{\partial z}, \quad H_\theta^{n*} = \alpha\delta t \frac{\partial \theta^{n*}}{\partial z}, \quad H_\pi^{n*} = \alpha\delta t \frac{\partial \Pi^{n*}}{\partial z}. \tag{A.3}$$

The divergence term D^{n*} in (2.15) or simply D in the Helmholtz equation (2.18), is given by:

$$D^{n*} = 1 + \alpha\delta t \gamma^{-1} \nabla \cdot \boldsymbol{u}^{n*}. \tag{A.4}$$

Appendix B. The complete coupled Helmholtz problem

Here we derive a more detailed representation of the combined Helmholtz problem. As a first step we replace (3.4) with the more concise equation

$$\boldsymbol{u}' + \boldsymbol{\mathcal{H}}_u(\Pi') + \boldsymbol{\mathcal{S}}(F') = \boldsymbol{\mathcal{R}}_u - \boldsymbol{\mathcal{S}} F^{n*} = \hat{\boldsymbol{\mathcal{R}}}_u, \tag{B.1}$$

where $\boldsymbol{\mathcal{S}}$ is the linear operator mapping the source distribution to the velocity vector; i.e. it represents the sum in (3.4) and induces the increments to $\mathcal{S}_u$, $\mathcal{S}_v$ and $\mathcal{S}_w$ to u, v and w respectively. Substituting this expression into (2.15) now gives

$$\mathcal{L}(\Pi') - H_\pi \overline{\mathcal{S}_w(F')} - \mathcal{H}_d[\boldsymbol{\mathcal{S}}(F')] = \mathcal{R}_\pi - \mathcal{H}_d(\hat{\boldsymbol{\mathcal{R}}}_u) - H_\pi \overline{\hat{\mathcal{R}}_w}, \tag{B.2}$$

from this we identify $\boldsymbol{S}$ in (3.11) as the discrete representation of

$$\hat{\mathcal{S}}[F'] = -H_\pi \overline{\mathcal{S}_w(F')} - \mathcal{H}_d[\boldsymbol{\mathcal{S}}(F')]. \tag{B.3}$$

Appendix C. The discrete Dirac delta function

In this paper the simplest form of the discrete delta function was used. This is defined using the locally compact function

$$\psi(x,h) = \begin{cases} 1 - |x/h|, & |x| < h, \\ 0, & |x| \geq h. \end{cases} \tag{C.1}$$

The discrete, 3 dimensional, delta function is then defined as, for $\boldsymbol{x} = (x, y, z)$,

$$\delta_h(\boldsymbol{x}) = \psi(x, \delta x)\psi(y, \delta y)\psi(z, \delta z), \tag{C.2}$$

where δx, δy and δz are the grid spacing for the simulation.

References

[1] T. Gal-Chen, R.C. Somerville, On the use of a coordinate transformation for the solution of the Navier-Stokes equations, J. Comput. Phys. 17 (2) (1975) 209–228.
[2] T. Allen, Comparison of iterative pressure solvers for turbulent flow over hills, Atmos. Sci. Lett. 8 (2007) 21–28.
[3] P.K. Smolarkiewicz, R. Sharman, J. Weil, S.G. Perry, D. Heist, G. Bowker, Building resolving large-eddy simulations and comparison with wind tunnel experiments, J. Comput. Phys. 227 (1) (2007) 633–653.
[4] N. Wood, A. Staniforth, A. White, T. Allen, M. Diamantakis, M. Gross, T. Melvin, C. Smith, S. Vosper, M. Zerroukat, J. Thuburn, An inherently mass-conserving semi-implicit semi-lagrangian discretization of the deep-atmosphere global non-hydrostatic equations, Q. J. R. Meteorol. Soc. 140 (682) (2014) 1505–1520.
[5] C.S. Peskin, Flow Patterns Around Heart Valves: a Digital Computer Method for Solving the Equations of Motion, Ph.D. thesis, The Albert Einstein College of Medicine, 1972.
[6] C.S. Peskin, The immersed boundary method, Acta Numer. 11 (2002) 479–517.
[7] R.P. Beyer, R.J. LeVeque, Analysis of a one-dimensional model for the immersed boundary method, SIAM J. Numer. Anal. 29 (2) (1992) 332–364.
[8] Y.-H. Tseng, J.H. Ferziger, A ghost-cell immersed boundary method for flow in complex geometry, J. Comput. Phys. 192 (2) (2003) 593–623.
[9] E. Fadlun, R. Verzicco, P. Orlandi, J. Mohd-Yusof, Combined immersed-boundary finite-difference methods for three-dimensional complex flow simulations, J. Comput. Phys. 161 (1) (2000) 35–60.
[10] R. Mittal, G. Iaccarino, Immersed boundary methods, Annu. Rev. Fluid Mech. 37 (2005) 239–261.
[11] D. Goldstein, R. Handler, L. Sirovich, Modeling a no-slip boundary condition with an external force field, J. Comput. Phys. 105 (1993) 354–366.
[12] M.-C. Lai, C.S. Peskin, An immersed boundary method with formal second-order accuracy and reduced numerical viscosity, J. Comput. Phys. 160 (2) (2000) 705–719.
[13] K. Lundquist, F. Katopodes Chow, J. Lundquist, An immersed boundary method for the weather research and forecasting model, Mon. Weather Rev. 138 (2010) 796–817.
[14] J. Bao, F.K. Chow, K.A. Lundquist, Large-eddy simulation over complex terrain using an improved immersed boundary method in the weather research and forecasting model, Mon. Weather Rev. 146 (9) (2018) 2781–2797.
[15] M. Diebold, C. Higgins, J. Fang, A. Bechmann, M.B. Parlange, Flow over hills: a large-eddy simulation of the Bolund case, Bound.-Layer Meteorol. 148 (1) (2013) 177–194.
[16] S. Chester, C. Meneveau, M.B. Parlange, Modeling turbulent flow over fractal trees with renormalized numerical simulation, J. Comput. Phys. 225 (1) (2007) 427–448.
[17] R. DeLeon, M. Sandusky, I. Senocak, Simulations of turbulent flow over complex terrain using an immersed-boundary method, Bound.-Layer Meteorol. 167 (3) (2018) 399–420.
[18] T. Allen, M. Zerroukat, A deep non-hydrostatic compressible atmospheric model on a Yin-Yang grid, J. Comput. Phys. 319 (2016) 44–60.
[19] R. Bisplinghoff, H. Ashley, R. Halfman, Aeroelasticity, Dover Publications, 1996.
[20] N. Muskhelishvili, Singular Integral Equations, Dover Publications, 1992.
[21] J.H. Ferziger, M. Peric, Computational Methods for Fluid Mechanics, 3rd edition, Springer, 2004.
[22] S. Eisenstat, H. Elman, M. Schultz, Variational iterative methods for non- symmetric systems of linear equations, SIAM J. Numer. Anal. 20 (1983) 345–357.
[23] G.H. Golub, C.F.V. Loan, Matrix Computations, 3rd edition, Johns Hopkins University Press, 1996.
[24] A. Rizzi, H. Viviand, Numerical Methods for the Computation of Inviscid Transonic Flows with Shocks, Notes on Numerical Fluid Mechanics, vol. 3, Vieweg, Braunschweig, 1981.
[25] I. Demirdžić, Ž. Lilek, M. Perić, A collocated finite volume method for predicting flows at all speeds, Int. J. Numer. Methods Fluids 16 (1993) 1029–1050.
[26] J.M. Straka, R.B. Wilhelmson, L.J. Wicker, J.R. Anderson, K.K. Droegemeier, Numerical solutions of a non-linear density current: a benchmark solution and comparisons, Int. J. Numer. Methods Fluids 17 (1) (1993) 1–22.
[27] M. Zerroukat, T. Allen, A three-dimensional monotone and conservative semi-Lagrangian scheme (SLICE-3D) for transport problems, Q. J. R. Meteorol. Soc. 138 (2012) 1640–1651.
[28] M. Zerroukat, T. Allen, On the monotonic and conservative transport on overset/Yin-Yang grids, J. Comput. Phys. 302 (2015) 285–299.